Xiangyu Wang

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Education

University of Illinois Urbana-Champaign | Fall 2023 – present

Ph.D. in Mathematics, in progress. Advisor: Xiaochun Li.

Research Interests

Harmonic analysis (Kakeya-type problems, Fourier restriction, decoupling, and local smoothing); additive combinatorics; analytic number theory.

Publications and Preprints

1. **Restriction Estimates Using Multilinear Kakeya** arXiv:2610.17752

   For sufficiently large dimensions, our approach yields better exponents than those obtained by a direct application of polynomial partitioning.

2. **Cone Restriction Estimates under Half-Scale Dimensional Conditions** arXiv:2610.16294

   Building on the framework of Du–Li–Wang–Zhang, we incorporate the broad restriction estimate of Ou–Wang and a localized multilinear estimate obtained from cyclic bilinear pairing and Kakeya summation of packet energies.

3. **Wave Packet Density in Local Smoothing: Refinements and Limitations** arXiv:2609.34675

   This work refines the wave packet density framework to improve local smoothing estimates for wave equations in all even spatial dimensions $n \geq 4$, including $p > 2.8862\ldots$ when $n = 4$. It also constructs an explicit Euclidean counterexample that reveals limitations of the existing framework.

4. **A Weighted Restriction Estimate in $\mathbb{R}^2$** arXiv:2609.12587

   With Zhenbin Cao, Feilong Guo, and Junfeng Li.

5. **On Oscillatory Integral Operators satisfying the Cinematic Curvature Condition** arXiv:2512.22516

6. **On Cone Restriction Estimates in Higher Dimensions** arXiv:2411.02961

   Accepted by Journal of Fourier Analysis and Applications.

7. **On Integers whose Friable Component is Large** arXiv:2609.37507

   With Gérald Tenenbaum.

8. **A Short Proof of a Theorem of Pan and Zhang** arXiv:2609.37508

Teaching Experience

University of Illinois Urbana-Champaign

Scheduled to lead discussion sections in Spring 2027.

Mathematics Tutor: MATH 241 (Calculus III). Provided homework help and answered students' questions.

Grader: MATH 512 (Algebraic Geometry), MATH 444 (Elementary Real Analysis), and MATH 448 (Complex Variables).